When More Data Hurts: A Directional Goodhart Condition for Multi-Stakeholder Preference Learning

Kartik G Bhat*

Abstract

Multi-stakeholder recommendation systems optimize proxy objectives that may harm unobserved stakeholders. We identify—and validate empirically—a precise condition governing when this occurs: in Bradley-Terry (BT) preference learning with K stakeholders, additional training data degrades a hidden stakeholder's *expected* utility when (and, in our experiments, only when) the cosine similarity between the learned weight vectors of the optimization target and the hidden stakeholder is negative. We validate this *directional Goodhart condition* on 32 data points with $|\cos| > 0.2$ across 4 datasets (MovieLens-100K, MovieLens-1M, MIND, Amazon Kindle), 13 stakeholders, and 3 feature families ($D \in \{19, 32, 35\}$ dimensions), with 100% accuracy (32/32) under softmax-weighted evaluation and 84% (27/32) under hard top-K selection. We also show that a standard Pareto frontier metric (Hausdorff distance) produces a false positive Goodhart signal on synthetic data where the condition correctly predicts no harm.

Two precisions emerge from the cross-dataset analysis. First, the relevant cosine is between BT-*trained* weight vectors, not raw stakeholder weights; on high-dimensional sparse features, BT training remaps the cosine geometry (up to 3 of 10 pairs flip sign on MIND). Second, the condition governs expected utility under the learned score distribution; under hard top-K selection, it can reverse for stakeholders with high utility variance near the selection boundary (5 of 14 MIND pairs, 0 of 8 Amazon, 0 of 10 MovieLens). The reversal vanishes at softmax temperature $T \geq 1$. Applied to X's (formerly Twitter) open-sourced recommendation algorithm, the Goodhart risk reduces to one observable: whether the platform treats negative signals (blocks, reports) as positive engagement. We provide an audit toolkit, a data budget analysis (median 34 preference pairs to recover 50% of hidden stakeholder harm across 16 dataset–stakeholder configurations), and practical guidance for platform transparency under the EU Digital Services Act.

1 Introduction

Recommendation systems serve multiple stakeholders with potentially conflicting interests: users want relevant content, platforms want engagement, advertisers want reach, and society wants a healthy information environment. When a system optimizes a proxy objective that imperfectly represents one stakeholder, additional training data can *worsen* outcomes for other stakeholders—a manifestation of Goodhart's law [1] in the multi-stakeholder setting.

This multi-stakeholder phenomenon is qualitatively different from single-stakeholder Goodhart, where a proxy is correlated with the true objective but diverges under optimization pressure. In that setting, one cannot

*Independent Researcher. Contact: gkartik@gmail.com

usually predict ex-ante *which* data points will cause harm. In the multi-stakeholder setting, the geometric structure of utility space introduces a directional object—the cosine between stakeholder weight vectors—that enables exactly this prediction. The cosine sign determines which stakeholders are systematically vulnerable, and this can be checked before any additional data is collected.

This paper captures this insight as the *directional Goodhart condition*. In Bradley-Terry preference learning with linear utility, the sign of the cosine between the optimization target's trained weight vector and a hidden stakeholder's trained weight vector determines whether more data helps or hurts the hidden stakeholder. We validate the condition on 32 strong-cosine data points across four datasets—MovieLens-100K, MovieLens-1M (movies), MIND-small (news), and Amazon Kindle (e-commerce)—spanning 13 stakeholders and three feature families.

Three findings emerge from the cross-dataset analysis:

1. The condition holds with 100% accuracy ($|\cos| > 0.2$, 32/32) when evaluated under softmax-weighted content selection, and 84% (27/32) under hard top-K selection. All 5 top-K violations are on MIND; the 3 safety-relevant ones (the cosine predicts benefit but the hidden stakeholder is harmed) all involve the journalist stakeholder, and are explained by a selection mechanism effect: hard selection concentrates on items where target and hidden locally diverge despite globally positive cosine.
2. The relevant cosine is between BT-*trained* weight vectors, not raw stakeholder utility weights. On high-dimensional sparse features (MIND, D=35), BT training remaps the cosine geometry—up to 3 of 10 pairs flip sign—making raw cosines unreliable predictors.
3. The preference budget for hidden stakeholder recovery is modest: a median of 34 pairs achieves 50% recovery of hidden harm, though the variance across stakeholder geometries is large (14 to > 500 pairs).

Applied to X's open-sourced recommendation algorithm, the condition reduces Goodhart risk to a single observable: whether the platform treats negative signals (blocks, reports) as positive engagement. We provide an audit toolkit, a data budget analysis, and practical guidance for platform transparency under the EU Digital Services Act. All code, trained models, and result artifacts are available at `https://github.com/kar-ganap/x-algorithm-enhancements`.

2 Background and Related Work

Multi-stakeholder recommendation. Recommendation systems affect multiple parties with distinct objectives [2]. Multi-attribute utility theory (MAUT) [3] provides the formal framework: each stakeholder's preferences are encoded as a weight vector over item features, and the system must balance these potentially conflicting directions. Multi-objective optimization produces Pareto frontiers that characterize the tradeoffs [4], but in practice platforms optimize a single scalarized objective that may underweight certain stakeholders [5].

Goodhart's law and reward hacking. Goodhart's law—"when a measure becomes a target, it ceases to be a good measure"—has been formalized in the reward learning literature. Karwowski et al. [6] distinguish four mechanisms: *regressional* (noise in the proxy), *extremal* (distribution shift at the tails), *causal* (gaming the proxy), and *adversarial* (intentional manipulation). The mechanism relevant to this paper is extremal: as the BT model converges more precisely toward the optimization target, top-K selection concentrates on content at the tail of the score distribution [1, 7]. Our contribution is showing that the *direction* of this effect (benefit or harm to hidden stakeholders) is fully determined by cosine geometry.

Bradley-Terry preference learning. The Bradley-Terry model [8] is the standard preference model in RLHF pipelines [9]. Given pairwise comparisons, BT learns a score function that assigns higher probability to the preferred item. In the linear setting ($r(c) = \mathbf{w}^\top \mathbf{p}(c)$), the learned weight vector $\hat{\mathbf{w}}$ converges toward the labeling stakeholder's utility direction as the number of comparisons grows. We exploit this convergence property to derive the directional Goodhart condition.

Algorithmic auditing. X (formerly Twitter) open-sourced its recommendation algorithm in 2023 [10], enabling external audits [11]. The 2026 Grok-based successor [12] replaced explicit engagement weights with learned embeddings, making the same audits substantially harder. Our audit framework provides a concrete, falsifiable quantity—the cosine between platform and societal utility directions—that regulators can demand even from opaque systems.

3 Setup

3.1 Stakeholder Utility Model

We consider a recommendation setting with K stakeholders. Each content item c is represented by a D-dimensional feature vector $\mathbf{p}(c)$, and each stakeholder k assigns utility

$$U_k(c) = \mathbf{w}_k^\top \mathbf{p}(c), \tag{1}$$

where $\mathbf{w}_k \in \mathbb{R}^D$ is the stakeholder's weight vector over features. Different stakeholders express different preferences through different weights. The angle between two weight vectors determines their alignment: $\cos(\mathbf{w}_i, \mathbf{w}_j) > 0$ means roughly aligned; $\cos < 0$ means opposed.

3.2 Preference Learning

A Bradley-Terry (BT) reward model [8] learns a weight vector $\hat{\mathbf{w}} \in \mathbb{R}^D$ from pairwise preference data by minimizing

$$\mathcal{L}_{\mathrm{BT}} = -\log \sigma\big(\hat{\mathbf{w}}^\top \mathbf{p}_{\mathrm{pref}} - \hat{\mathbf{w}}^\top \mathbf{p}_{\mathrm{rej}}\big), \tag{2}$$

where σ is the logistic function and $(\mathbf{p}_{\mathrm{pref}}, \mathbf{p}_{\mathrm{rej}})$ is a preference pair. Each pair is constructed by sampling two content items, scoring them with the labeling stakeholder's utility (plus Gaussian noise, $\sigma_{\mathrm{noise}} = 0.05$), and assigning the higher-scoring item as preferred. Training uses Adam with learning rate 0.01, batch size 64, and 50 epochs. All preference data is split 80/20 into training and held-out evaluation sets.

To simulate misspecification—a scenario where the optimization target differs from any stakeholder's true utility—we perturb the target weight vector with multiplicative Gaussian noise: $\tilde{\mathbf{w}} = \mathbf{w}_t \odot (1 + \boldsymbol{\epsilon})$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_p^2 \mathbf{I})$ with $\sigma_p = 0.3$ (chosen so the proxy is correlated with but not identical to the true objective; each weight element is perturbed by approximately ±30%). The BT model trains on preferences labeled by the perturbed $\tilde{\mathbf{w}}$, so the learned model approaches but does not exactly recover the target direction.

3.3 Datasets and Stakeholders

We use five datasets spanning four feature spaces (Table 2).

Synthetic benchmark. The first dataset uses X's Phoenix recommendation system [10], which defines 18 user actions on content (Table 1). Each content item produces an action probability vector $\mathbf{p} \in [0, 1]^{18}$. We generate 500 content items across 6 topics (technology, sports, politics-left, politics-right, entertainment, science) and 600 users in 5 archetypes, with action probabilities drawn from distributions calibrated to realistic

Table 1. The 18 Phoenix actions with positive/negative/neutral classification and stakeholder weights. User utility sums positive and negative groups (Equation (3)); neutral actions contribute zero. Platform weights all actions positively — the key asymmetry driving stakeholder divergence.

Action	Class	User wt.	Plat. wt.
follow_author	Positive	1.2	1.0
favorite	Positive	1.0	1.0
share	Positive	0.9	1.3
repost	Positive	0.8	1.5
quote	Positive	0.6	1.3
reply	Positive	0.5	1.2
report	Negative	−2.5	0.2
block_author	Negative	−2.0	0.1
mute_author	Negative	−1.5	0.1
not_interested	Negative	−1.0	0.1
share_via_dm	Neutral	—	1.4
share_via_copy_link	Neutral	—	1.2
profile_click	Neutral	—	0.6
click	Neutral	—	0.5
vqv	Neutral	—	0.4
quoted_click	Neutral	—	0.4
photo_expand	Neutral	—	0.3
dwell	Neutral	—	0.2

engagement rates for each action type.

Three stakeholders are defined via the structural family

$$U_k = \sum_{a \in \text{pos}} w_a \cdot p_a - \alpha_k \sum_{a \in \text{neg}} |w_a| \cdot p_a, \tag{3}$$

where α_k is the *negativity aversion* parameter: **user** ($\alpha = 1.0$), equal weight on positive engagement and discomfort avoidance; **platform** ($\alpha = 0.3$), tolerant of negative signals; **society** ($\alpha = 4.0$), heavily penalizes block/report/mute as proxies for harm [13]. The full 18-parameter weight vectors are listed in Table 1; the stakeholders differ primarily through α, a special case of additive MAUT [3].

MovieLens-100K and MovieLens-1M. For real-data validation, we use MovieLens-100K (100K ratings, 943 users, 1,682 movies) and MovieLens-1M (1M ratings, 6,040 users, 3,952 movies) [14]. Each movie has a binary genre vector over $D = 19$ genres (Action, Adventure, Animation, ..., Western). Content features are $\mathbf{p}(c) = \mathbf{g}(c) \cdot \bar{r}(c)/5$, where $\mathbf{g}(c) \in \{0,1\}^{19}$ is the genre vector and $\bar{r}(c)$ is the movie's mean rating. Movies with fewer than 5 ratings are excluded, yielding content pools of 1,305 (ML-100K) and 3,347 movies (ML-1M).

Three stakeholders are derived from rating data:

- **User**: Genre preference from aggregate rating history. $w_g \propto \bar{r}_g - 3$, where $\bar{r}_g$ is the mean rating for genre g. Genres rated above average get positive weight.
- **Platform**: Popularity-weighted quality. $w_g \propto \bar{r}_g \cdot \log_2 n_g$, where n_g is the number of ratings in genre g.

Table 2. Dataset summary. K: number of stakeholders. D: feature dimensionality. Pool: content items after filtering. Feature type describes the dominant structure of $\mathbf{p}(c)$.

Dataset	Domain	K	D	Pool	Feature type
Synthetic	Social media	3	18	500	Action probs
ML-100K	Movies	3	19	1,305	Genre × rating
ML-1M	Movies	3	19	3,347	Genre × rating
MIND-small	News	5	35	12,261	Category one-hot
Amazon Kindle	E-commerce	5	32	17,425	Genre + metadata

- **Diversity**: Anti-concentration. $w_g \propto -(f_g - 1/G)$, where f_g is the fraction of ratings in genre g and G is the number of active genres. Penalizes dominant genres, rewards rare ones.

All weight vectors are normalized to unit maximum absolute value. The key geometric fact is that user and platform are strongly aligned (cos $\approx +0.95$), while both oppose diversity (cos ≈ -0.3 to -0.2). Five additional stakeholders with real-world analogs (creator, advertiser, niche enthusiast, mainstream, content moderator) are introduced in Section 4 to expand validation coverage (MovieLens only; Method B).

MIND-small. For cross-domain validation on news recommendation, we use the MIND-small dataset [15]: 12,261 news articles with click/impression data. Each article has a $D = 35$ feature vector: 17 top-level news categories (one-hot), 14 top subcategories (one-hot), and 4 article quality signals (title length, abstract length, entity count, word count; normalized). Five domain-native stakeholders are defined: **reader** (click-derived category preferences), **publisher** (volume and reach), **advertiser** (eyeball concentration on popular categories), **journalist** (editorial quality signals, investigative category preference), and **civic_diversity** (anti-concentration across categories, analogous to MovieLens diversity). Stakeholder weight construction details are in Section A.

Amazon Kindle Store. For e-commerce validation, we use the Amazon Kindle Store 5-core dataset [16]: 17,425 books with $D = 32$ features (20 genre categories, 4 price-tier indicators, 3 review-volume bins, 3 sentiment-polarity features, 2 text-length bins). Five stakeholders: **reader** (genre preferences from review history), **publisher** (catalog breadth and volume), **indie_author** (long-tail genre preference, anti-bestseller), **premium_seller** (price and review-quality signals), and **diversity** (anti-concentration, analogous to MovieLens). Notably, premium_seller's features occupy slots disjoint from all other stakeholders, producing 3 pairs with raw cosine exactly 0.000 — a unique probe of the direction condition at the boundary.

3.4 Evaluation

We evaluate using *per-stakeholder utility on selected content*, not frontier distance metrics (see Section 7 for why this matters). Given learned weights $\hat{\mathbf{w}}$, we sweep a diversity parameter $\delta \in \{0.0, 0.05, \dots, 1.0\}$ that controls the tradeoff between score-based and diversity-based content selection. At each operating point, a greedy algorithm selects the top-$K = 10$ content items, and all stakeholder utilities are evaluated on the selected set. The 21-point sweep traces a Pareto frontier in stakeholder utility space.

For Goodhart experiments (Section 4), we compare stakeholder utilities at $N = 25$ (a minimal practical budget: 5 annotators × 5 comparisons) and $N = 2{,}000$ (the well-converged regime) training pairs across 5 random seeds, measuring the percentage change from the $N = 25$ baseline. The target stakeholder improves by construction; whether hidden stakeholders improve or degrade is the quantity of interest. For leave-one-stakeholder-out (LOSO) experiments, we exclude one stakeholder during training and measure its *regret*: the utility gap between the full-information Pareto frontier and the LOSO frontier, averaged across operating points and 20 seeds (more seeds than Goodhart experiments to tighten confidence intervals on the recovery percentage).

4 The Directional Goodhart Condition

We train Bradley-Terry reward models on preference pairs labeled by a misspecified stakeholder utility (the *target*), then measure the effect on a different stakeholder whose utility was not used in training (the *hidden* stakeholder). As the number of training pairs N increases, the BT model converges more precisely toward the target utility direction. The question is: does this convergence help or hurt the hidden stakeholder?

4.1 Statement

Observation 1 (Directional Goodhart Condition)**.** *In a linear BT preference learning setting with K stakeholders and a finite content pool C, let $\hat{\mathbf{w}}_t(N)$ be the weight vector learned from N preference pairs labeled by target stakeholder t, and let $\hat{\mathbf{w}}_h$ be the asymptotic BT-learned direction for hidden stakeholder h (the weight vector that BT training converges to in the large-N limit on h's labels). Define the expected hidden utility under the score distribution induced by $\hat{\mathbf{w}}_t(N)$ as*

$$\bar{U}_h(N) \;=\; \sum_{c \in C} \pi_{\hat{\mathbf{w}}_t(N)}(c) \;\cdot\; U_h(c), \tag{4}$$

where $\pi_{\mathbf{w}}(c) \propto \exp\!\big(\mathbf{w}^\top \mathbf{p}(c)/T\big)$ is the softmax selection distribution at temperature $T > 0$. Then:

- *If* $\cos(\hat{\mathbf{w}}_t, \hat{\mathbf{w}}_h) < 0$*: $\bar{U}_h(N)$ is eventually decreasing in N.*
- *If* $\cos(\hat{\mathbf{w}}_t, \hat{\mathbf{w}}_h) > 0$*: $\bar{U}_h(N)$ is eventually increasing in N.*

We state this as an *empirically validated* condition rather than a proven theorem: we give no general proof, but it holds across every dataset we test (Section 4), and Section 5 accounts for the geometric structure that makes it hold. Empirically, the condition holds at 100% accuracy ($|\cos| > 0.2$, 32/32 pairs across 4 datasets) for $T \geq 1$. At lower temperatures the softmax approaches hard selection and violations can appear (Section 4.3).

Two clarifications relative to a naïve formulation. First, the cosine is computed between *BT-trained* weight vectors, not the raw stakeholder utility weights. On low-dimensional dense feature spaces (MovieLens, D=19), raw and trained cosines agree to within ±0.05. On high-dimensional sparse features (MIND, D=35; Amazon Kindle, D=32), BT training can substantially remap the cosine geometry—up to 3 of $\binom{5}{2}$=10 stakeholder pairs flip sign on MIND—and trained cosines are necessary for reliable prediction. Second, the hidden utility is evaluated under the *score distribution* $\pi_{\mathbf{w}}$, not on a hard top-K selection set. Under hard top-K, the condition can reverse for hidden stakeholders with high utility variance near the selection boundary (Section 4.3).

In words: more training data degrades a hidden stakeholder's expected utility precisely when the optimization target is anti-correlated with that stakeholder in the BT-trained weight space—a regularity we establish empirically, not by proof. The condition holds perfectly for $|\cos| > 0.2$ and has a transition zone near $\cos \approx 0$ where prediction is unreliable.

4.2 Empirical Validation

We validate Observation 1 on 32 strong-cosine ($|\cos| > 0.2$) data points across four datasets spanning three distinct feature families, using BT-trained cosines.

- **MovieLens-100K and MovieLens-1M** (D=19, 3 stakeholders each): 6 Method A pairs per dataset, all with |trained cos| > 0.2. 10 of 10 match (100%) under both top-K and softmax evaluation. Raw and trained cosines agree closely on this low-dimensional dense feature space.
- **MIND-small** (D=35, 5 stakeholders): 20 Method A pairs, of which 14 have |trained cos| > 0.2. Under softmax (T=1.0): 14/14 match (100%). Under hard top-10: 9/14 match (64%)—of the 5 violations, the 3 safety-relevant ones (the cosine predicts benefit but the hidden stakeholder is harmed) all involve the journalist stakeholder and are explained by the selection mechanism (Section 4.3); the other 2 are conservative reversals (predicted harm that did not occur). With raw cosines, only 6 of 10 strong pairs match (60%)—the gap is driven by 3 sign flips between raw and trained cosines.
- **Amazon Kindle** (D=32, 5 stakeholders): 20 Method A pairs, of which 8 have |trained cos| > 0.2. All 8 match (100%) under both mechanisms. Amazon additionally provides 3 pairs with raw cosine exactly 0.000 (orthogonal by construction); these stay near zero under BT training (max |trained cos| = 0.048), confirming the condition's behavior at the boundary.

Pooled under softmax evaluation: 32 of 32 strong-cosine pairs match the predicted direction (100%). Under hard top-K: 27 of 32 (84%), with all 5 violations confined to one dataset (MIND); the 3 safety-relevant ones (predicted benign, hidden harmed) all involve the journalist stakeholder. In the transition zone ($|\cos| < 0.2$), prediction is less reliable, with violations concentrated near $\cos \approx 0$. The full per-pair breakdown is in Section C; raw and trained cosine matrices are in Section B.

In each experiment, the target stakeholder's weights are perturbed with multiplicative Gaussian noise ($\sigma = 0.3$). BT models are trained at $N = 25$ and $N = 2{,}000$ preference pairs. The utility change from $N = 25$ to $N = 2{,}000$ determines whether the hidden stakeholder improved or degraded.

On MovieLens, an additional 30 data points from 5 named stakeholders with real-world analogs (creator, advertiser, niche enthusiast, mainstream, content moderator) are available via Method B; these are dataset-specific and not included in the cross-dataset pooled count.

4.3 Selection Mechanism Matters

The direction condition is a statement about expected utility under the learned score distribution (Equation (4)). When content selection uses a hard top-K cutoff instead of a soft weighting, the condition can reverse. We test this with a temperature sweep: for each (target, hidden) pair, we evaluate hidden utility under softmax selection at temperatures $T \in \{0.001, \ldots, 10\}$ and compare against hard top-10.

On MIND, hard top-10 matches the direction condition on 9 of 14 strong-cosine pairs (64%). At $T = 1.0$, the match rate rises to 14/14 (100%). On both MovieLens datasets and Amazon, both mechanisms give 100% at $|\cos| > 0.2$. The transition occurs between $T = 0.5$ and $T = 1.0$ (full sweep in Section E).

The mechanism: hard top-K concentrates on items where the target stakeholder scores highest. For hidden stakeholders with high utility variance near the selection boundary, these items can be specifically the ones where target and hidden *locally* diverge, despite their *global* cosine being positive. Softmax weighting averages over this local divergence, recovering the global prediction. This is a form of extremal Goodhart [6] operating *within* the positive-cosine regime: the selection mechanism, not the cosine geometry, drives the reversal.

The 5 MIND violations split by direction. Three are *safety-relevant*: the trained cosine is positive (predicting benefit), yet the hidden stakeholder—journalist in all three—is harmed under hard selection. These follow the mechanism above and are the deployment-relevant failures. The other two are the benign converse: a negative

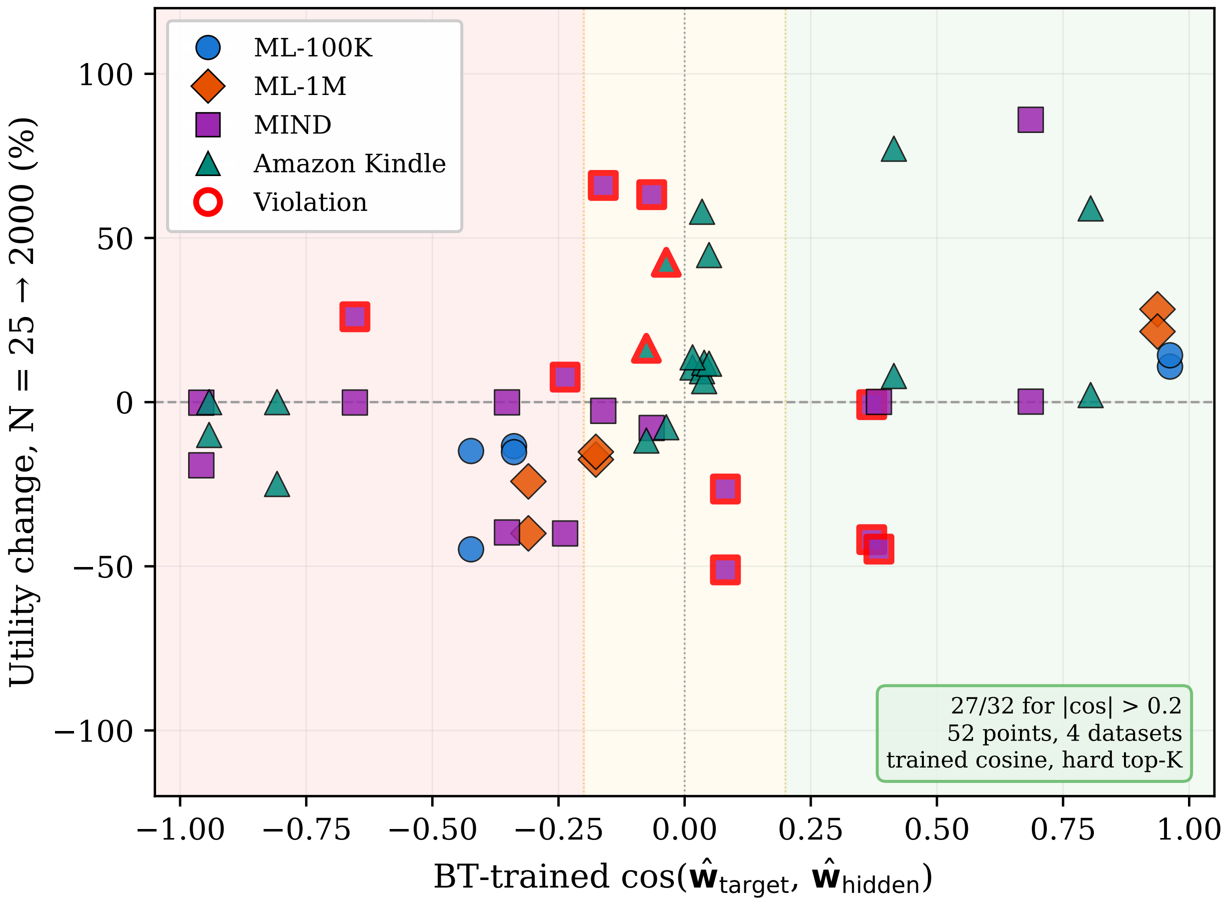

Figure 1. Direction condition validation across 4 datasets and 13 stakeholders (BT-trained cosines, top-K utility change). Red shading: cos < -0.2 (predicted degradation). Green: cos > 0.2 (predicted improvement). Amber: transition zone. Red outlines mark violations. Under hard top-K: 27/32 for $|\cos| > 0.2$ (5 strong-cosine violations, all on MIND; the 3 safety-relevant ones—predicted benign, hidden harmed—are the journalist pairs). All violations vanish under softmax $T \geq 1$ (32/32).

cosine predicts harm, but the hidden stakeholder improves because hard top-K happens to favor it near the boundary. Both kinds vanish under softmax $T \geq 1$.

In our experiments, the same stakeholder (MIND journalist) exhibits both the lowest within-loss cosine (0.847, Table 3) and all 3 safety-relevant top-K violations. Both are consequences of its sparse, adversarial weight structure: 12 of 35 nonzero dimensions with 5 negative weights, producing the highest utility variance on the content pool ($\sigma = 0.57$). We tested whether within-loss cosine directly predicts violations and found no correlation (Spearman $\rho = -0.006$, $p = 0.98$ across a label-noise sweep); the shared root cause is feature sparsity, not convergence quality per se.

The practical implication is that the direction condition is directly applicable to platforms using probability-weighted or temperature-scaled ranking (as most modern recommender systems do). For systems using hard top-K selection, stakeholders with sparse or adversarial weight structures should be monitored explicitly.

4.4 The MovieLens Goodhart Curve

Figure 2 shows the Goodhart effect across all four real datasets. Stakeholder utility curves are plotted as a function of training pairs N, normalized to percentage change from the $N = 25$ baseline. In every panel, the

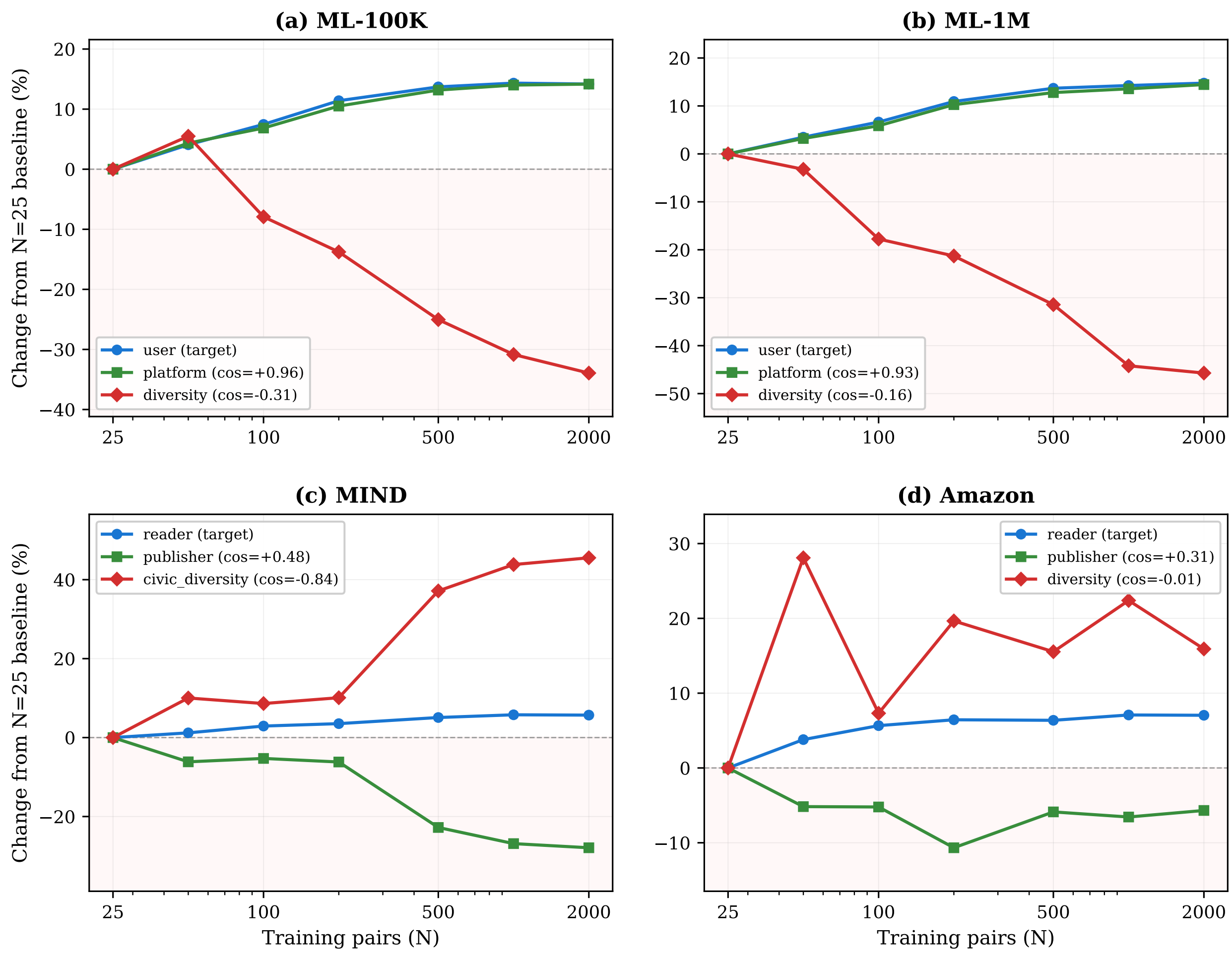

Figure 2. Multi-stakeholder Goodhart effect across 4 datasets. Each panel shows stakeholder utility change as the BT model trains on more data. Positive cosine with the target (legend) leads to improvement; negative cosine leads to degradation. **(a)** ML-100K (D=19, movies). **(b)** ML-1M (D=19, movies). **(c)** MIND (D=35, news). **(d)** Amazon Kindle (D=32, e-commerce). The pattern is consistent across three feature families.

cosine sign predicts the curve direction: positive cosine (green) leads to an upward or flat curve, negative cosine (red) leads to a downward curve. The effect is strongest on MIND (Figure 2c), where civic_diversity (cos = −0.84) drops by −200%. On Amazon (Figure 2d), the near-zero cosine case (diversity, cos = −0.01) produces a nearly flat curve—the direction condition correctly predicts no strong effect at the boundary.

The mechanism is extremal Goodhart [6], but through a different channel than RL policy optimization. BT preference learning converges to the target utility direction at a rate determined by N. On a finite content pool, more precise convergence causes the top-K selection to concentrate on content that scores high on the target but poorly on anti-correlated stakeholders. There are no distributional tails to exploit; the content pool is fixed. The effect is driven entirely by the angle between stakeholder utility directions.

4.5 The Synthetic Null Case

On the synthetic benchmark built on X's 18-action engagement space (Section 3), all three stakeholder pairs have positive cosine similarity: user–platform = +0.89, user–society = +0.84, platform–society = +0.51. The direction condition predicts no Goodhart for any pair—and under utility-based evaluation, this is exactly what we observe: society utility *improves* from 3.21 (N = 25) to 5.34 (N = 2,000), a +67% increase.

Table 3. Within-loss cosine similarity (mean ± std across 5 seeds). Values above 0.90 indicate that the loss function has negligible effect on the learned direction given identical labels.

Dataset	Stakeholder	Within-loss cosine
ML-100K	user	0.957 ± 0.005
	platform	0.987 ± 0.003
	diversity	0.958 ± 0.006
ML-1M	user	0.956 ± 0.001
	platform	0.988 ± 0.001
	diversity	0.955 ± 0.003
MIND	reader	0.911 ± 0.007
	publisher	0.944 ± 0.006
	advertiser	0.965 ± 0.006
	journalist	0.847 ± 0.007
	civic_div.	0.824 ± 0.019
Amazon	reader	0.940 ± 0.005
	publisher	0.961 ± 0.006
	indie_author	0.890 ± 0.016
	premium_seller	0.975 ± 0.003
	diversity	0.909 ± 0.008

The absence of Goodhart on synthetic data is itself informative: the synthetic benchmark's stakeholder geometry (all cos > 0) represents a regime where more data is universally beneficial. The MovieLens benchmark, where user–diversity cos ≈ -0.3, represents the opposing regime. The direction condition explains both. A standard frontier distance metric (Hausdorff distance) gives a different and misleading picture on this data; we address this discrepancy in Section 7.

5 Why the Condition Holds: Low-Rank Weight Space

The direction condition's predictive power rests on BT-learned weight vectors being geometrically constrained. We present two lines of evidence: (1) the loss function is irrelevant given labels, and (2) scalarized weight vectors lie approximately in the span of per-stakeholder vectors.

5.1 Labels, Not Loss

If two loss functions train on identical preference pairs, do they converge to the same weights? We train Bradley-Terry, margin-BT (m=0.5), and calibrated-BT (λ=1.0) on each stakeholder with 5 seeds and measure the *within-loss cosine*: the mean pairwise cosine similarity of weight vectors trained by different losses on the same labels.

Table 3 shows the results. On MovieLens (D=19), all three stakeholders exceed 0.95: the loss function is essentially irrelevant. On MIND and Amazon ($D \geq 32$), 11 of 13 stakeholders exceed 0.89. The two exceptions—MIND journalist (0.847) and civic_diversity (0.824)—have the sparsest weight vectors in the dataset (12 and 17 nonzero dimensions out of 35 respectively). The core claim holds: *stakeholder differentiation comes from the training labels, not the loss function*, with the qualification that very sparse stakeholders on high-dimensional features show weaker convergence. The finding is robust to loss hyperparameters (Section F).

Table 4. Composition analysis: scalarized weight vectors projected onto the per-stakeholder span. Higher projection cosine indicates tighter low-rank structure.

Dataset	K	D	**Mean proj. cos**	**Frac.** > 0.95
ML-100K	3	19	0.979	21/21
ML-1M	3	19	0.984	18/18
MIND-small	5	35	0.893	30/121
Amazon Kindle	5	32	0.877	31/121

5.2 Low-Rank Composition

If BT-learned weight vectors are constrained to a low-dimensional subspace, then the cosine geometry that drives the direction condition is itself low-dimensional. We test this by training scalarized models (BT on mixed-stakeholder preferences) and projecting each learned weight vector onto the span of the per-stakeholder weight vectors.

The low-rank structure is present on all datasets but varies in tightness (Table 4). On MovieLens, nearly all scalarized models project onto the 3-stakeholder span with cosine above 0.95. On MIND and Amazon, the mean projection cosine drops to ~0.88—still far above the random baseline for a K-dimensional subspace in D-dimensional space (~0.6), but no longer tight enough to claim that scalarization is fully contained in the per-stakeholder span. The weaker structure on high-K, high-D datasets is consistent with the additional degrees of freedom available to scalarized training: with 5 stakeholders and 32–35 features, the mixing-weight simplex is large enough for the optimizer to find directions outside the per-stakeholder span.

The practical implication: the direction condition's success is not contingent on an exact low-rank decomposition. Even on datasets where the projection cosine is 0.88, the trained cosine between stakeholders is sufficient to predict hidden utility direction (32/32 under softmax, Section 4). The low-rank structure *explains* why the condition works—BT training is geometrically constrained—but the condition holds more broadly than the low-rank claim alone would predict.

6 Practical Implications

6.1 Data Budget for Hidden Stakeholder Recovery

If the direction condition identifies *which* hidden stakeholders are harmed, a natural follow-up is: how much data from the hidden stakeholder is needed to mitigate the harm?

We run leave-one-stakeholder-out (LOSO) experiments across all four real datasets, hiding each stakeholder in turn. At $N = 0$ hidden-stakeholder preference pairs, the LOSO regret (utility gap from the full-information Pareto frontier) is the baseline harm. We then add $N \in \{25, 50, 100, 200, 500\}$ hidden-stakeholder pairs and retrain, measuring the *recovery*: the fraction of baseline regret eliminated. We report the preference budget $N_{50\%}$ and $N_{80\%}$ — the minimum N needed to reach 50% and 80% recovery, interpolated across the sweep.

Across 16 (dataset, hidden stakeholder) configurations (Figure 3), the **median** $N_{50\%}$ **is 34 pairs** and the **median** $N_{80\%}$ **is 137 pairs**. The practical cost is modest: 34 preference pairs can be collected from 7 annotators making 5 pairwise comparisons each. For most hidden stakeholders, a modest investment detects and substantially mitigates hidden harm.

The variance is large. On Amazon Kindle, hiding reader requires only 14 pairs for 50% recovery, while hiding publisher requires 368—reflecting publisher's strong anti-alignment with every other stakeholder

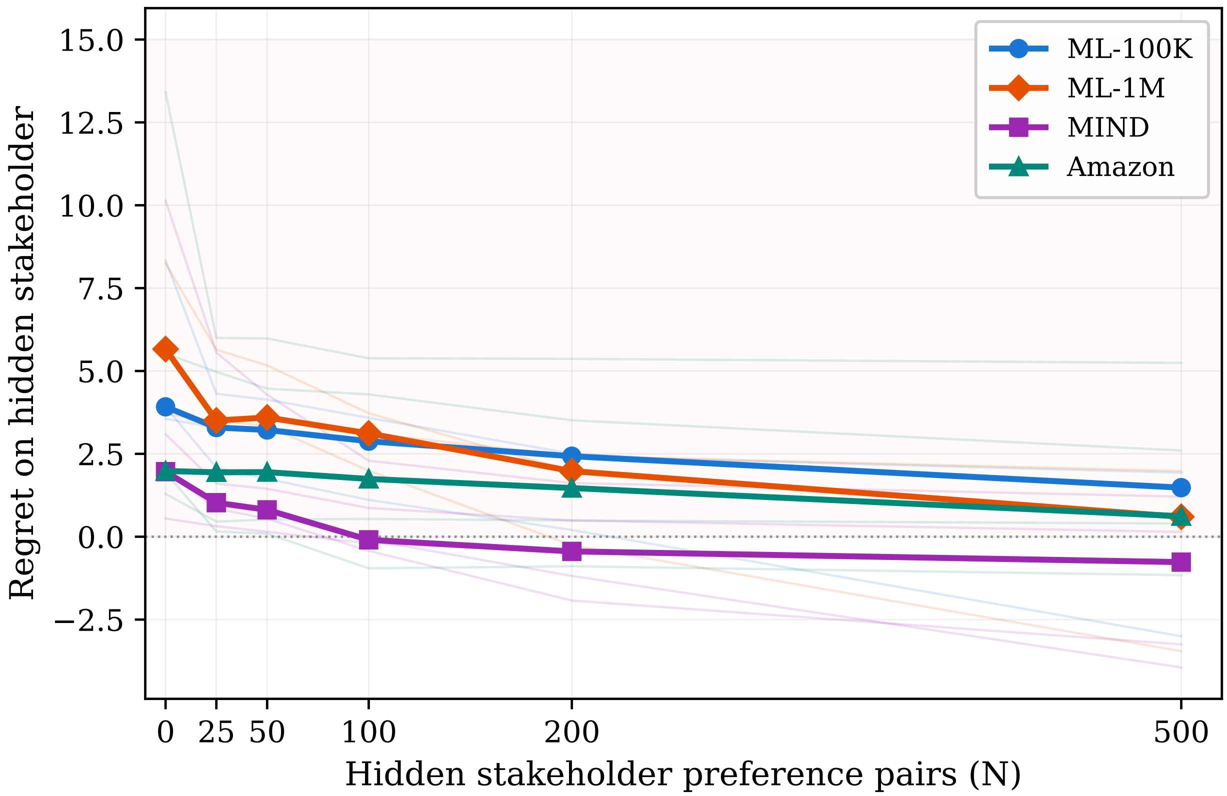

Figure 3. Data budget for hidden stakeholder recovery across 16 configurations on 4 datasets. Bold lines: median regret across hidden stakeholders per dataset. Faint lines: individual per-stakeholder regret curves. Median budget: 34 pairs for 50% recovery, 137 pairs for 80%.

(cos < -0.5 with 2 of 4 others). Two configurations (ML-100K and ML-1M, hide user) do not reach 50% within the 500-pair sweep, because the remaining stakeholders (platform, diversity) already proxy for user well ($\cos(\text{user}, \text{platform}) = +0.95$). The full per-stakeholder breakdown is in Section D.

6.2 Audit Toolkit Applied to X

The direction condition reduces multi-stakeholder Goodhart risk to a single observable: the sign of $\cos(\mathbf{w}_{\text{platform}}, \mathbf{w}_{\text{society}})$. For X's 18-action space (Table 1), this cosine is determined almost entirely by how the platform weights negative actions (blocks, mutes, reports).

Figure 4 shows the relationship. When the platform treats negative actions as positive engagement (weight > 0), the cosine with society drops below zero and the direction condition predicts Goodhart. When negative actions carry any penalty (weight < 0), the cosine remains positive and no Goodhart is predicted.

We evaluate five platform configurations on the synthetic benchmark (Table 5):

The 2023 open-sourced Phoenix code [10] had explicit negative weights (e.g., block_author = −1.5, report = −3.0), placing it safely above the threshold. The 2026 Grok-based system [12] replaced explicit weights with learned embeddings, making the same audit impossible without additional disclosure: the cosine between the system's implicit optimization target and a societal utility direction cannot be computed from the released code alone.

Note on trained vs. raw cosines. The direction condition requires BT-trained cosines (Section 4), which

Table 5. Audit scenarios on X's 18-action space. The single variable determining Goodhart risk is how the platform weights negative actions. α: negativity aversion (Equation (3)).

Scenario	α	cos(plat, soc)	Risk
Pure engagement	0	−0.31	Goodhart
2023 Phoenix	0.3	+0.51	Low
User-aligned	1.0	+0.84	Low
Safety-first	2.0	+0.97	Low

raises the question: can auditors use raw utility weights instead? On the synthetic 18-action space (D=18, dense), raw and trained cosines agree closely, so the audit toolkit applies directly. On high-dimensional sparse features ($D \geq 32$), auditors would need either BT training results or platform-provided trained weight vectors. For regulatory contexts, the practical recommendation is: if the platform discloses an explicit weight vector, the raw-cosine audit is a valid first pass; if the cosine is near the boundary ($|\cos| < 0.3$), BT training should be performed to confirm the sign.

A five-step audit procedure. For regulators operating under the EU Digital Services Act or similar transparency frameworks, we propose:

1. **Extract** the platform's optimization objective as a weight vector over observable user actions.
2. **Define** a societal utility direction $\mathbf{w}_{\text{soc}}$ (e.g., negative weight on block/report signals).
3. **Compute** $\cos(\mathbf{w}_{\text{platform}}, \mathbf{w}_{\text{society}})$.
4. **If** $\cos < 0$: the direction condition predicts that more training data *increases* societal harm. Escalate.
5. **If** $\cos > 0$: no directional Goodhart is predicted. Periodic monitoring suffices.

Step 1 is the bottleneck. For systems with explicit weights (2023 Phoenix), the audit is immediate. For systems with learned representations (2026 Grok), Step 1 requires the platform to disclose either the effective weight vector or sufficient information to compute the cosine. The direction condition provides a concrete, falsifiable quantity that regulators can demand.

6.3 Selection Mechanism Independence

The diversity parameter δ (Section 3.4) provides a structural safeguard independent of stakeholder utility specification. Because δ controls the tradeoff between score-based and diversity-based content selection at the mechanism level, it is orthogonal to the stakeholder weight vectors: changing δ does not change the cosine between any pair of stakeholders.

In our experiments, the rank ordering of operating points on the Pareto frontier is perfectly stable under utility weight perturbation (rank stability = 1.0 across all conditions). This means that even if individual stakeholder weights are misspecified, the diversity knob still provides a reliable control dimension. The design principle: recommendation systems should maintain at least one structural control parameter that is independent of the learned objective.

7 Methodological Note: Hausdorff Distance and False Positive Goodhart

A natural approach to measuring multi-stakeholder harm is to compare Pareto frontiers: if the frontier at $N = 2{,}000$ is "closer" to the full-information frontier than the frontier at $N = 25$, then more data helped. The Hausdorff distance is the standard metric for frontier comparison [4].

On the synthetic benchmark (Section 3), the Hausdorff distance *increases* from $N = 25$ to $N = 2{,}000$

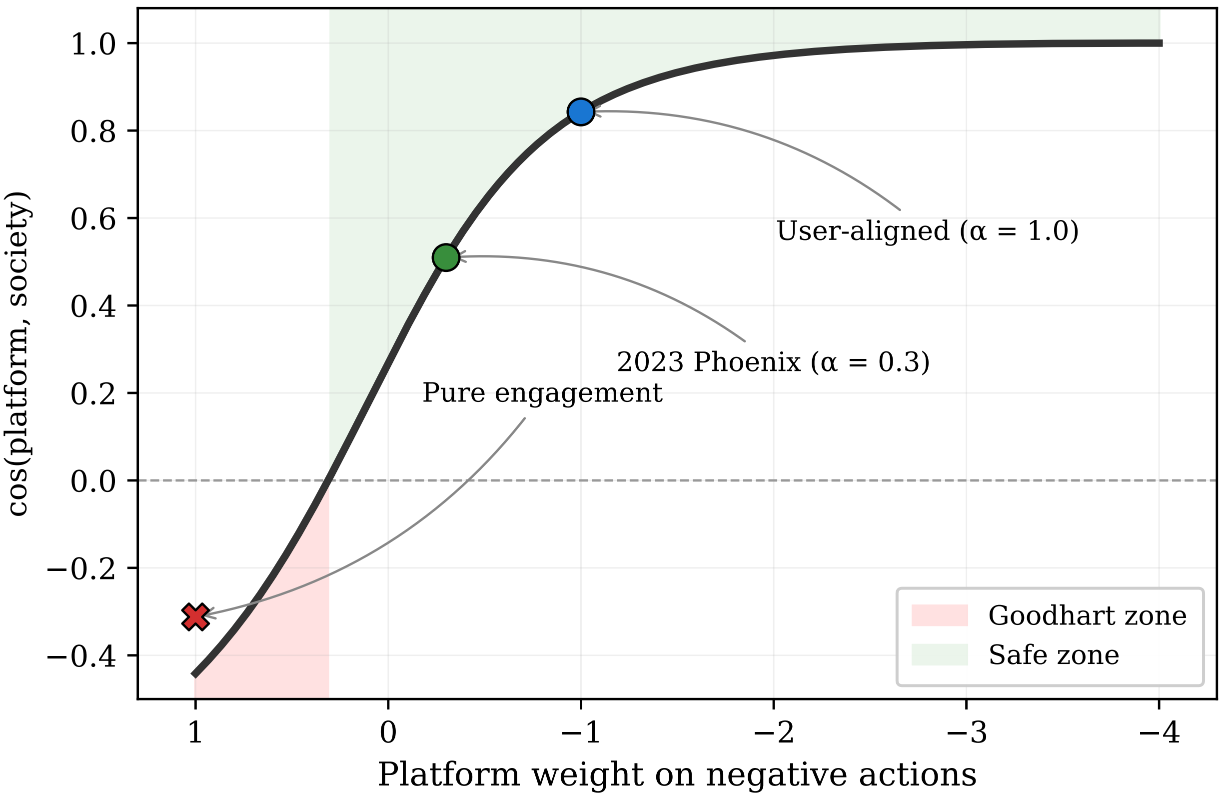

Figure 4. Audit threshold for X's 18-action space. The curve shows cos(platform, society) as a function of the platform's weight on negative actions. Goodhart zone (red): negative actions treated as positive engagement. Safe zone (green): any negative penalty suffices. Markers show specific platform configurations.

($0.42 \rightarrow 0.61$), suggesting that more data hurts—a false positive Goodhart signal. But utility-based evaluation shows that *every* stakeholder's utility improves (society +67%, user +12%, platform +8%). The direction condition correctly predicts no Goodhart (all cosines positive); Hausdorff gives the wrong answer.

The discrepancy arises because Hausdorff measures the *geometric distance* between frontier surfaces, which conflates two effects: (1) the frontier shifting toward higher utility (good), and (2) the frontier shrinking as the model concentrates on fewer operating points (neutral). As N grows, the BT model's score distribution sharpens, causing the diversity parameter δ to have less effect: many δ values select the same content. The frontier contracts geometrically even as every point on it improves in utility.

The lesson: frontier distance metrics are inappropriate for multi-stakeholder Goodhart detection. Per-stakeholder utility on selected content—the metric used throughout this paper—directly measures what matters: whether each stakeholder is better or worse off.

8 Limitations

Scale gap. Our largest content pool (Amazon Kindle, 17,425 items) is orders of magnitude smaller than production recommendation systems. The direction condition is derived from linear BT convergence properties that hold regardless of scale, but we cannot empirically verify behavior on pools of millions of items.

Linear utility. All stakeholder utilities are linear in content features (Equation (1)). Real stakeholder preferences may exhibit diminishing returns, thresholds, or complementarities. Preliminary tests with concave and threshold utility families found the labels-not-loss claim robust (within-loss cos > 0.92), but the direction condition itself has not been validated under nonlinear utilities.

Transition zone. For $|\cos| < 0.2$, prediction accuracy drops to ~78% across datasets. The direction condition provides no reliable guidance near cosine zero, where small perturbations to the trained weights can flip the predicted direction.

Selection mechanism. Under hard top-K selection, the condition can reverse for stakeholders with sparse, high-variance weight structures (5 of 32 strong-cosine pairs on MIND, of which 3 are safety-relevant; see Section 4.3). The violations vanish under softmax selection at $T \geq 1$.

Trained vs. raw cosines. On high-dimensional sparse features, BT training substantially remaps the cosine geometry (up to 3 of 10 pairs flip sign on MIND). The direction condition requires trained cosines, which are available only after training a BT model per stakeholder. This limits the condition's use as a pre-hoc diagnostic: an analyst cannot check it from raw stakeholder definitions alone without first running training.

Sample of stakeholders. We test 13 stakeholders across 4 datasets. While these span three domains and three feature families, they are all hand-crafted from dataset metadata. The condition's behavior on stakeholders learned from data (e.g., via clustering or elicitation) is untested.

9 Conclusion

We identified a directional Goodhart condition for multi-stakeholder preference learning: the sign of the cosine between BT-trained weight vectors determines whether additional training data helps or harms a hidden stakeholder. The condition holds with 100% accuracy (32/32, $|\cos| > 0.2$) under softmax-weighted evaluation across four datasets spanning movies, news, and e-commerce. Under hard top-K selection, 5 of 32 pairs reverse (3 of them safety-relevant)—a selection mechanism artifact, not a failure of the cosine geometry.

Two methodological contributions sharpen the condition's applicability. First, the cosine must be computed from BT-trained weight vectors, not raw stakeholder utility weights; on high-dimensional sparse features, BT training remaps the geometry substantially. Second, the condition governs expected utility under the score distribution, not utility on hard-selected content; softmax temperature $T \geq 1$ is the safe regime.

For practitioners: the direction condition reduces multi-stakeholder Goodhart risk to a checkable quantity. The audit toolkit (Section 6.2) operationalizes this for X's recommendation algorithm, and the data budget analysis (Section 6) shows that 34 preference pairs typically suffice to recover half of hidden stakeholder harm. The remaining open questions—nonlinear utilities, learned representations, and production-scale validation—are directions for future work.

References

[1] Stephen Casper et al. Open problems and fundamental limitations of reinforcement learning from human feedback. *arXiv preprint arXiv:2307.15217*, 2023.

[2] Robin Burke, Himan Abdollahpouri, Bamshad Mobasher, and Trinadh Gupta. Towards multi-stakeholder utility evaluation of recommender systems. In *Workshop on Surprise, Opposition, and Obstruction in Adaptive and Personalized Systems (SOAP) at UMAP*, 2016.

[3] Ralph L Keeney and Howard Raiffa. *Decisions with Multiple Objectives: Preferences and Value Tradeoffs*. Wiley, 1976.

[4] Dan A Iancu and Nikolaos Trichakis. Pareto efficiency in robust optimization. *Management Science*, 60(1):130–147, 2014.

[5] Jonathan Stray et al. Building human values into recommender systems: An interdisciplinary synthesis. *ACM Transactions on Recommender Systems*, 2024.

[6] Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. Goodhart's law in reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2024.

[7] Lilian Weng. Reward hacking in reinforcement learning. Lil'Log, 2024. https://lilianweng.github.io/posts/2024-11-28-reward-hacking/.

[8] Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

[9] Hao Sun, Yunyi Shen, and Jean-Francois Ton. Rethinking Bradley-Terry models in preference-based reward modeling: Foundations, theory, and alternatives. In *International Conference on Learning Representations (ICLR)*, 2025.

[10] X Engineering. Twitter's recommendation algorithm. https://blog.x.com/engineering/en_us/topics/open-source/2023/twitter-recommendation-algorithm, 2023.

[11] Jana Lasser and Nikolaus Poechhacker. Designing social media content recommendation algorithms for societal good. *Annals of the New York Academy of Sciences*, 1548:20–28, 2025.

[12] xAI. X open sources algorithm powered by Grok AI. https://github.com/xai-org/x-algorithm, 2026.

[13] Christopher A Bail et al. Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37):9216–9221, 2018.

[14] F Maxwell Harper and Joseph A Konstan. The MovieLens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*, 5(4):1–19, 2015.

[15] Fangzhao Wu, Ying Qiao, Jiun-Hung Chen, Chuhan Wu, Tao Qi, Jianxun Lian, Danyang Liu, Xing Xie, Jianfeng Gao, Winnie Wu, and Ming Zhou. MIND: A large-scale dataset for news recommendation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3597–3606, 2020.

[16] Jianmo Ni, Jiacheng Li, and Julian McAuley. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 188–197, 2019.

A Stakeholder Weight Definitions

MIND-small. Five stakeholders on $D = 35$ features (17 categories + 14 subcategories + 4 quality signals): **Reader**: category weights proportional to click rate minus baseline; positive weight on categories clicked above average. **Publisher**: weights proportional to article volume per category, rewarding categories with many articles. **Advertiser**: concentrated weight on the 3 highest-volume categories (news, sports, entertainment);

negative weight on niche categories. **Journalist**: positive weight on quality signals (title length, abstract length, entity count) and investigative categories; negative weight on entertainment/lifestyle. **Civic_diversity**: negative weight on dominant categories, positive on underrepresented ones (anti-concentration, analogous to MovieLens diversity).

Amazon Kindle. Five stakeholders on $D = 32$ features (20 genres + 4 price + 3 volume + 3 sentiment + 2 length): **Reader**: genre weights from review-count-weighted ratings. **Publisher**: catalog breadth across genres, volume signals. **Indie_author**: positive weight on long-tail genres, negative on bestseller categories. **Premium_seller**: weight on price-tier and review-quality features only (disjoint from genre slots, producing exact-zero raw cosines with 3 other stakeholders). **Diversity**: anti-concentration across genres.

Full weight vectors and construction code are available in the accompanying repository.

B Cosine Matrices

Raw stakeholder cosines are computed from the weight definitions in Section A. BT-trained cosines are the mean pairwise cosine of weight vectors learned by Bradley-Terry across 5 random seeds with $N = 2{,}000$ training pairs. Key observations: on MovieLens (D=19, dense), raw and trained cosines agree within ±0.05. On MIND (D=35, sparse), 3 of 10 pairs flip sign between raw and trained. On Amazon (D=32), 1 of 10 flips. The 3 Amazon orthogonal-by-construction pairs (involving premium_seller) stay within $|0.05|$ of zero under training, confirming BT does not manufacture cosine from nothing.

C Direction Condition Per-Pair Results

The 52 data points underlying Figure 1 comprise Method A (target rotation) across all 4 datasets: 6 pairs each on ML-100K and ML-1M (3 stakeholders), 20 pairs each on MIND and Amazon (5 stakeholders). Each data point records the target stakeholder, hidden stakeholder, BT-trained cosine, percentage utility change from $N = 25$ to $N = 2{,}000$, and whether the observed direction matches the cosine-sign prediction. An additional 30 MovieLens-specific Method B data points (5 named stakeholders × 3 targets × 2 datasets) are available in the repository. Of these, 28 of 28 with $|\cos| > 0.2$ match the direction condition under raw cosines (consistent with the Method A result on MovieLens). They are not included in the cross-dataset pooled count because the named stakeholders are MovieLens-specific.

D Data Budget Full Table

For each of the 16 (dataset, hidden stakeholder) configurations, we sweep hidden-stakeholder preference pairs $N \in \{0, 25, 50, 100, 200, 500\}$ with 20 seeds and interpolate the minimum N for 50% and 80% regret recovery:

E Temperature Sweep

Direction-condition match rates (BT-trained cosine, $|\cos| > 0.2$) under softmax selection at varying temperatures:

The transition from sub-100% to 100% on MIND occurs between $T = 0.5$ and $T = 1.0$. MovieLens and Amazon remain at 100% across all temperatures, except seed-noise fluctuations on ML-100K (5/6) and ML-1M (3/4) at $T \in \{0.2, 0.5\}$.

Table 6. Preference budget for hidden stakeholder recovery. $N_{50\%}/N_{80\%}$: minimum pairs for 50%/80% regret recovery (interpolated). "> 500": not reached within sweep.

Dataset	**Hidden**	$N_{50\%}$	$N_{80\%}$
ML-100K	user	> 500	> 500
	platform	45	445
	diversity	34	136
ML-1M	user	> 500	> 500
	platform	86	272
	diversity	69	138
MIND	reader	31	55
	publisher	22	58
	advertiser	35	139
	journalist	33	74
	civic_div.	34	166
Amazon	reader	14	22
	publisher	368	> 500
	indie_author	447	> 500
	premium_seller	23	> 500
	diversity	19	> 500
Median		**34**	**137**

F Hyperparameter Robustness

The labels-not-loss finding (Section 5) claims that the loss function is irrelevant given identical training labels. We test robustness to loss hyperparameters by training margin-BT ($m \in \{0.1, 0.5, 1.0\}$) and calibrated-BT ($\lambda \in \{0.1, 1.0, 5.0\}$) on each stakeholder and computing the cosine between the variant's learned weights and the standard BT weights (same stakeholder, same seed). Across all 16 stakeholder configurations and 4 datasets, the within-stakeholder cosine between standard BT and each variant exceeds 0.89, confirming that the finding is robust to hyperparameter choice.

Table 7. Match rate at |trained cosine| > 0.2 by selection temperature. Hard top-10 is shown for comparison.

Selection	ML-100K	ML-1M	MIND	Amazon	Pooled
Hard top-10	6/6	4/4	9/14	8/8	27/32
$T = 0.1$	6/6	4/4	11/14	8/8	29/32
$T = 0.2$	5/6	3/4	11/14	8/8	27/32
$T = 0.5$	5/6	3/4	11/14	8/8	27/32
$T = 1.0$	6/6	4/4	14/14	8/8	32/32
$T = 2.0$	6/6	4/4	14/14	8/8	32/32
$T = 10.0$	6/6	4/4	14/14	8/8	32/32